

Muhammad Rayan Atta

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Summary of Qualifications

- Electrical Engineering Co-op student available for a 12-month work term beginning May 2027, with hands-on experience taking analog circuit designs from schematic through layout, simulation and verification.
- Designed and laid out a three-stage MOSFET amplifier and a four-layer sensor PCB, using SPICE and KiCad to move from schematic capture through component-level layout and design-rule verification.
- Compared calculated, simulated and measured circuit behaviour to debug and refine analog designs, iterating through structured, repeatable testing.
- Clear communicator experienced presenting technical work to project teams and translating design decisions into concise documentation for review.

Technical Skills

- Analog & Circuit Design:** Circuit analysis, analog electronics, MOSFET biasing topologies, small-signal AC analysis, DC operating-point calculations, gain/bandwidth trade-offs, electric networks, component selection, datasheet interpretation
- CAD & Layout Tools:** KiCad (schematic capture, PCB layout, netlist generation, ERC/DRC), SPICE, MultiSIM, Altera Quartus, AutoCAD/CAD concepts
- Digital Logic & Verification:** VHDL, FPGA synthesis and deployment, testbench simulation, waveform analysis, truth-table-driven verification
- Testing & Instrumentation:** Oscilloscope, digital multimeter, signal generator, bench power supplies; structured troubleshooting, test documentation
- Programming & Communication:** C; Python, C++, Java (working knowledge); Microsoft Excel, Word, PowerPoint and Outlook; technical presentations and cross-team communication

Education

Toronto Metropolitan University

Bachelor of Engineering in Electrical Engineering (Co-op), GPA: 3.64/4.33

Toronto, ON

Sept 2024 – June 2029 (Expected)

Honours: FEAS Dean's List (twice).

Relevant Coursework: Electric Networks, Circuit Analysis, Electronic Circuits I, Digital Systems

Technical Projects

Low-Power MOSFET Amplifier System

TMU | SPICE, Analog IC Design Concepts | Apr 2026

- Designed a three-stage common-source MOSFET amplifier topology under a 3.3 V supply and sub-1 mW power budget, balancing gain, bandwidth, biasing, loading and output-drive requirements across stages.
- Derived DC operating points through hand bias calculations, then ran AC small-signal SPICE simulations to validate gain, bandwidth and frequency response, debugging stage by stage until simulated results converged with calculated targets.

CanSat IMU & Magnetometer Breakout PCB

TMU CanSat | KiCad | Aug 2026

- Translated electrical and mechanical requirements into a four-layer, 3.3 V PCB, using datasheets to select sensors and develop the schematic, netlist and component-level layout, including a shared SPI interface.
- Applied layout considerations including trace routing and power/ground plane placement, closing the design with 0 DRC errors and 0 unconnected items; produced Gerber/drill files and documented trade-offs and verification results.

General-Purpose Processor (ALU)

TMU | VHDL, FPGA, Digital Logic Design | Dec 2025

- Designed an 8-bit ALU core in VHDL supporting 8 arithmetic and logical operations, integrated with FSM-based control logic and a 4x16 decoder, then synthesized and deployed the complete processor onto an FPGA development board.
- Verified functional correctness through testbench simulation, waveform analysis and truth-table-driven test cases, and produced structured documentation including truth tables, block/circuit diagrams and simulation waveforms.

GridSYNC AI Workload Peak Mitigation

TMU / Schneider Electric | Sept 2026

- Collaborated in a six-student team on an energy-management system concept combining flexible-load scheduling, battery storage sizing, peak shaving and fault recovery.
- Contributed research and technical content to a 19-slide presentation covering system architecture, CO2e calculations and a six-phase implementation roadmap.

Relevant Experience

Student Support Specialist - TMU Student Life & Learning Support

Sept 2026 – Present

- Serve as a frontline contact at the Student Learning Centre, responding to student questions and supporting Academic Accommodation Support administrators with intake and detail-sensitive administrative workflows.